INPT
المعهد الوطني للبريد والمواصلات
ⵜⴰⵙⴷⴰⵡⵜ ⵏ ⵜⵙⵍⵉⵜ ⴰⵏⴰⵎⵓⵔ ⵜⵉⵏⵙⵏⴰⵡⵉⵏ
Institut National des Postes et Télécommunications

PFE ENTERPRISE

End-of-Study Project Report

To obtain the Engineering Degree

Major: [Major / Program]

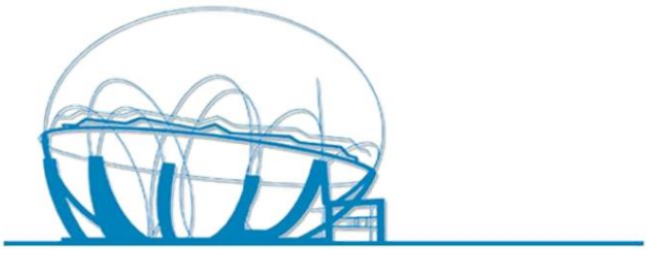
[Project Title]

Project carried out at PFE ENTERPRISE

Authored by:
Mr. ANOUAR EL IDRISSI

Defended on [date], before the jury composed of:

Mr./Ms. [Name]	[Institution]	- Examiner
Mr./Ms. [Name]	[Institution]	- Internal Supervisor
Mr./Ms. [Name]	PFE ENTERPRISE	- External Supervisor



Agence Nationale de Réglementation des Télécommunications

Institut National des Postes et Télécommunications

Academic year: [20XX/20XX]

Dedication

Replace this page with an optional personal dedication, or remove its input from `main.tex`.

Acknowledgements

I would like to thank the people and organizations that supported this end-of-study project. Replace this paragraph with your acknowledgements, including your academic and professional supervisors where appropriate.

Abstract

This report describes [*project title*], an end-of-study project completed at **PFE ENTERPRISE**. The project responds to [*brief problem statement*] by proposing [*solution*]. The work covers the analysis of requirements, the design and implementation of the solution, and its validation against defined criteria. Replace this sample with a concise overview of your context, method, contribution, and key result.

Keywords: project template, end-of-study project, requirements, design, implementation, validation.

Résumé

Ce rapport présente [*titre du projet*], un projet de fin d'études réalisé chez **PFE ENTERPRISE**. Le projet répond à [*problème*] en proposant [*solution*]. Le travail couvre l'analyse des besoins, la conception et l'implémentation de la solution, puis sa validation. Remplacez ce paragraphe par un résumé concis de votre contexte, méthode, contribution et résultat principal.

Mots-clés : modèle de rapport, projet de fin d'études, besoins, conception, implémentation, validation.

الملخص

يقدم هذا التقرير مشروع نهاية الدراسة بعنوان [Project Title] المنجز لدى PFE ENTERPRISE. استبدل هذا النص بملخص موجز يوضح سياق المشروع والمشكلة التي يعالجها والحل المقترح والمنهجية والنتائج الرئيسية.

الكلمات المفتاحية: مشروع نهاية الدراسة، قالب التقرير، التحليل، التصميم، التنفيذ، التحقق.

Contents

Acknowledgements **2**

Abstract **3**

Résumé **4**

Arabic Abstract **5**

Table of Contents **6**

List of Figures **7**

List of Tables **8**

List of Acronyms **9**

General Introduction **1**

1 Organization and Project Context **2**

1.1 Host Organization 2

1.2 Project Context 2

1.3 Problem Statement 2

1.4 Objectives and Scope 2

2 Requirements Analysis **3**

2.1 Stakeholders and Users 3

2.2 Functional Requirements 3

2.3 Non-functional Requirements 3

2.4 Planning and Methodology 4

3 Analysis and Design **5**

3.1 Proposed Solution 5

3.2 Architecture 5
3.3 Data and Process Design 5
3.4 Technology Choices 5

4 Implementation 6
4.1 Development Environment 6
4.2 Key Features 6
4.3 User Interface and Demonstration 6
4.4 Deployment 8

5 Validation and Results 9
5.1 Validation Strategy 9
5.2 Results 9
5.3 Discussion 9

Conclusion and Perspectives 11

Bibliography 12

Webography 14

List of Figures

4.1 Example architecture figure. Replace the image and caption with content relevant to your project. 7

List of Tables

2.1 Functional-requirements registry. Replace the sample rows with your own requirements. 3

5.1 Requirements-validation matrix. Replace the sample rows with your validation evidence. 9

List of Acronyms

Acronym	Meaning
API	Application Programming Interface
CSV	Comma-Separated Values
DB	Database
FR	Functional Requirement
NFR	Non-Functional Requirement
PDF	Portable Document Format
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience

Replace, add, or remove rows to match the terms used in your report. Keep the acronym in the left column and its full meaning in the right column.

General Introduction

This report presents the work completed during an end-of-study project at **PFE ENTERPRISE**. Replace this paragraph with the context of your organization, its sector, and the opportunity that motivated the project.

The project addresses [*a problem to solve*]. Explain why the problem matters, who is affected, and the limitations of the current situation. The objective is to design and deliver [*the proposed solution*]. State the principal objectives, expected deliverables, and boundaries of the work.

This report is organized as follows. Chapter 1 introduces the organization and project context. Chapter 2 describes the needs and requirements. Chapter 3 presents the analysis and design. Chapter 4 explains the implementation. Chapter 5 documents validation and results.

Chapter 1

Organization and Project Context

1.1 Host Organization

Introduce **PFE ENTERPRISE**: its activity, services, organization, and the team hosting the project. Include only information that helps the reader understand the project setting.

1.2 Project Context

Describe the business or technical context that led to the project. Explain the current process, product, or service and identify the opportunity for improvement.

1.3 Problem Statement

State the problem precisely. A useful formulation identifies the users, the current limitation, and the consequences of leaving it unresolved.

1.4 Objectives and Scope

List the main objective, secondary objectives, expected deliverables, and work outside the project scope. Replace this text with project-specific content.

Conclusion

This chapter establishes the context and objectives that guide the following requirements analysis.

Chapter 2

Requirements Analysis

2.1 Stakeholders and Users

Identify the people who use, operate, approve, or maintain the solution. Describe their needs in a short paragraph or a table.

2.2 Functional Requirements

Present the capabilities the solution must provide. For each requirement, add an identifier, description, priority, and acceptance criterion.

ID	Area	Requirement	Priority
FR-01	[*Feature area*]	The system shall allow a user to [*complete an action*].	High
FR-02	[*Feature area*]	The system shall display [*information*].	Medium
FR-03	[*Feature area*]	The system shall validate [*input or condition*].	High

Table 2.1: Functional-requirements registry. Replace the sample rows with your own requirements.

2.3 Non-functional Requirements

Document quality attributes such as performance, usability, security, reliability, maintainability, and accessibility when relevant.

2.4 Planning and Methodology

Explain the chosen work method and planning approach. Replace this sample with milestones, iterations, or a short schedule that reflects your project.

Conclusion

The validated requirements provide the basis for the system design.

Chapter 3

Analysis and Design

3.1 Proposed Solution

Present the solution at a high level. Describe how it addresses the problem stated in Chapter 1 and satisfies the requirements from Chapter 2.

3.2 Architecture

Explain the major components, their responsibilities, and the information exchanged between them. Insert an architecture diagram only when it clarifies the explanation.

3.3 Data and Process Design

Describe the principal data structures, user flows, interfaces, or business processes. Select the design views that are useful for your project.

3.4 Technology Choices

List the main technologies and briefly justify each choice according to the project constraints.

Conclusion

The design defines a clear implementation path for the proposed solution.

Chapter 4

Implementation

4.1 Development Environment

Describe the development environment, project organization, and configuration needed to reproduce the work.

4.2 Key Features

Present the most important implemented features. For each feature, explain the purpose, the implementation approach, and the result for the user.

4.3 User Interface and Demonstration

Add only the screenshots that help a reader understand a significant workflow. Each screenshot should have a readable caption and a short explanation in the surrounding text.

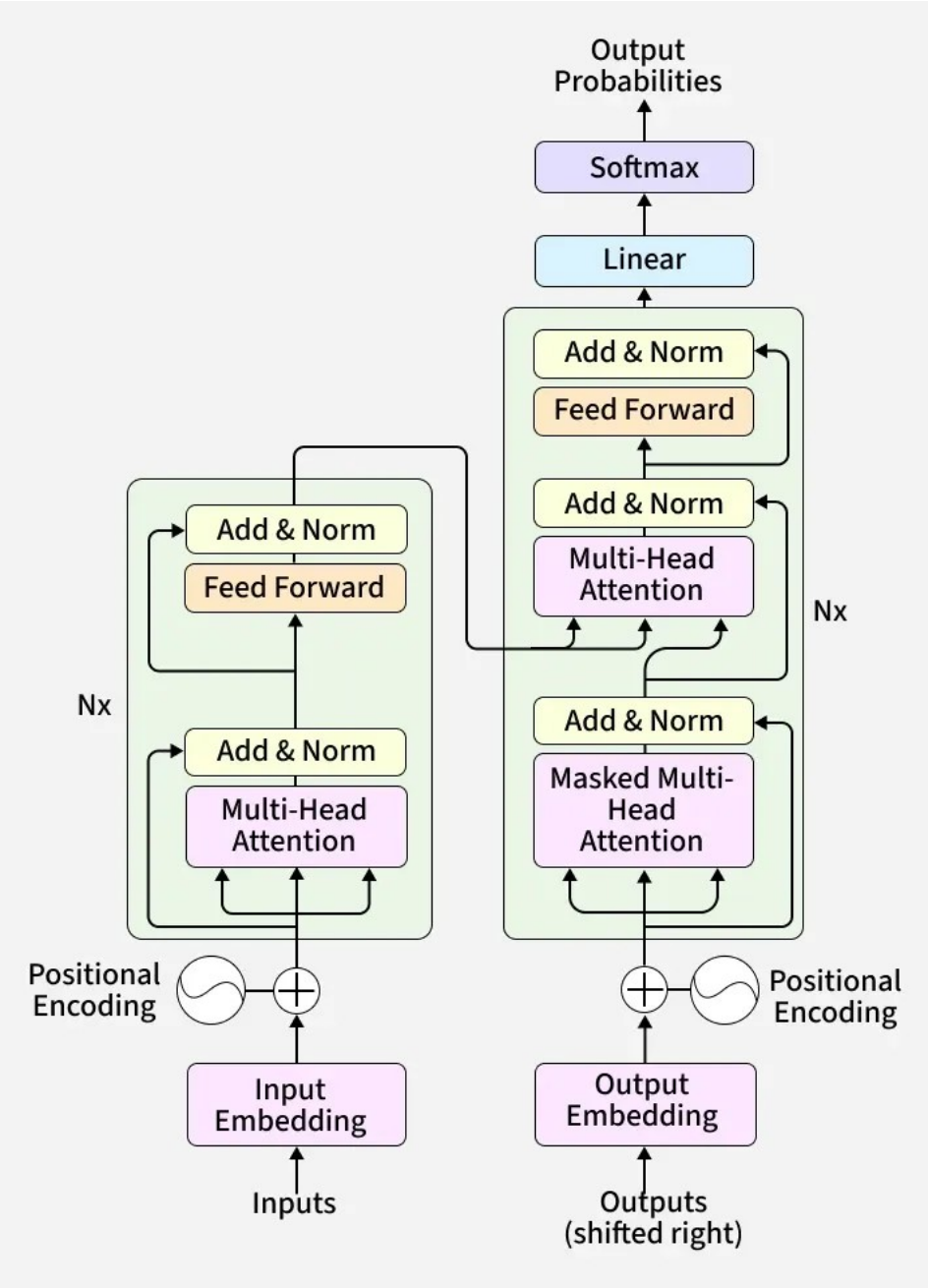


Figure 4.1: Example architecture figure. Replace the image and caption with content relevant to your project.

To insert an image, place it in a folder such as `Figures/` and replace the placeholder above with:

```
\includegraphics[width=0.78\textwidth]{Figures/your-image.png}
```

Use a clear filename, reference the figure in the text as Figure 4.1, and keep only images that support the explanation.

4.4 Deployment

Explain how the solution is delivered or deployed, if this is within the project scope.

Conclusion

This chapter shows how the proposed design was translated into a working solution.

Chapter 5

Validation and Results

5.1 Validation Strategy

Describe how the solution was validated. State the test scenarios, evaluation criteria, participants, or data used, as appropriate.

5.2 Results

Present the results using concise text, tables, or figures. Explain what each result demonstrates and avoid including raw output that does not support a conclusion.

ID	Validation target	Evidence or observed result	Status
V-01	[*Expected behavior or requirement*]	[*Test result, measure, or user feedback*]	Pass
V-02	[*Expected behavior or requirement*]	[*Test result, measure, or user feedback*]	Pass
V-03	[*Expected behavior or requirement*]	[*Test result, measure, or user feedback*]	Pending

Table 5.1: Requirements-validation matrix. Replace the sample rows with your validation evidence.

5.3 Discussion

Interpret the results in relation to the objectives and requirements. Mention observed limitations and the conditions under which the results apply.

Conclusion

Summarize whether the solution meets the defined objectives and prepare the closing conclusion.

Conclusion and Perspectives

This report presented [*project title*], completed at **PFE ENTERPRISE**. Replace this paragraph with a concise summary of the problem, the delivered solution, and the contribution of the project.

The project was validated through [*tests, demonstrations, measurements, or user feedback*]. Summarize the most important result and explain how it relates to the objectives defined at the start of the report.

Future work may include [*improvements, new features, wider deployment, or further study*]. Keep this section focused on realistic next steps.

Bibliography

Replace these sample entries with the books, papers, standards, and technical documentation cited in your report. Cite an entry in the text with `\cite{reference-key}`.

Bibliography

[1] Author Name, *Title of the Book*, Publisher, edition, year.

[2] Author Name, "Title of the Article," *Journal or Conference Name*, volume, issue, pages, year.

[3] Organization Name, *Title of the Standard or Technical Report*, reference number, year.

Webography

List the online sources used in the report. Include the organization or author, page title, URL, and access date.

Bibliography

[1] Organization Name, *Page Title*. Available at: https://example.com. Accessed on: [day month year].

[2] Organization Name, *Official Documentation*. Available at: https://example.com/documentation. Accessed on: [day month year].